

Date: 03/09/2025

Lab Practical #13:

To develop network using distance vector routing protocol and link state routing protocol.

Practical Assignment #13:

1. C/Java Program: Distance Vector Routing Algorithm using Bellman Ford's Algorithm.

```
import java.util.Scanner;

public class DistanceVectorRouting {
    private static final int INF = 9999;

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of routers: ");
        int numRouters = scanner.nextInt();

        int[][] costMatrix = new int[numRouters][numRouters];
        System.out.println("Enter the cost matrix (use " + INF + " for infinity):");
        for (int i = 0; i < numRouters; i++) {
            for (int j = 0; j < numRouters; j++) {
                costMatrix[i][j] = scanner.nextInt();
            }
        }

        int[][] distanceVector = new int[numRouters][numRouters];
        int[][] nextHop = new int[numRouters][numRouters];

        for (int i = 0; i < numRouters; i++) {
            for (int j = 0; j < numRouters; j++) {
                distanceVector[i][j] = costMatrix[i][j];
                nextHop[i][j] = (costMatrix[i][j] != INF && i != j) ? j : -1;
            }
        }
    }
}
```

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```
}

boolean updated;
do {
    updated = false;
    for (int i = 0; i < numRouters; i++) {
        for (int j = 0; j < numRouters; j++) {
            for (int k = 0; k < numRouters; k++) {
                if (distanceVector[i][k] + distanceVector[k][j] < distanceVector[i][j]) {
                    distanceVector[i][j] = distanceVector[i][k] + distanceVector[k][j];
                    nextHop[i][j] = nextHop[i][k];
                    updated = true;
                }
            }
        }
    }
} while (updated);

System.out.println("\nFinal Distance Vector Table:");
for (int i = 0; i < numRouters; i++) {
    System.out.println("Router " + (i + 1) + ":");
    for (int j = 0; j < numRouters; j++) {
        if (distanceVector[i][j] == INF) {
            System.out.print("INF ");
        } else {
            System.out.print((distanceVector[i][j] + 1) + " ");
        }
    }
    System.out.println();
}

scanner.close();
}
```

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}

2. C/Java Program: Link state routing algorithm.

```
import java.util.*;
public class Dijkstra {
    static final int INF = Integer.MAX_VALUE;
    static int findKey(boolean[] visited, int[] distance, int V) {
        int min = INF;
        int key = -1;

        for (int i = 0; i < V; i++) {
            if (!visited[i] && distance[i] < min) {
                min = distance[i];
                key = i;
            }
        }
        return key;
    }
    static void dijkstra(int[][] graph, int src) {
        int V = graph.length;
        boolean[] visited = new boolean[V];
        int[] distance = new int[V];
        Arrays.fill(distance, INF);

        distance[src] = 0;

        for (int i = 0; i < V - 1; i++) {
            int u = findKey(visited, distance, V);
            if (u == -1) break;

            visited[u] = true;

            for (int v = 0; v < V; v++) {
```

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```
        if (graph[u][v] != 0 && !visited[v] && distance[u] != INF
            && distance[v] > distance[u] + graph[u][v]) {
            distance[v] = distance[u] + graph[u][v];
        }
    }

    System.out.println("\nShortest distances from node " + src + ":");
    for (int i = 0; i < V; i++) {
        if (distance[i] == INF)
            System.out.println("Node " + i + ": INF");
        else
            System.out.println("Node " + i + ": " + distance[i]);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of vertices: ");
        int V = sc.nextInt();

        int[][] graph = new int[V][V];
        System.out.println("Enter adjacency matrix (0 if no edge):");
        for (int i = 0; i < V; i++) {
            for (int j = 0; j < V; j++) {
                graph[i][j] = sc.nextInt();
            }
        }

        System.out.print("Enter source node (0 to " + (V - 1) + "): ");
        int src = sc.nextInt();

        System.out.println("\nGraph:");
        for (int i = 0; i < V; i++) {
            for (int j = 0; j < V; j++)
                System.out.print(graph[i][j] + "\t");
            System.out.println();
        }
        dijkstra(graph, src);
    }
```



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}

}